```
# Install Hugging Face Transformers and required dependencies
!pip install transformers[sentencepiece] datasets evaluate accelerate
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
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  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.
```

```
# Import necessary libraries for summarization tasks
from transformers import pipeline
import torch
import pandas as pd

print("Transformers library imported successfully!")
print(f"PyTorch version: {torch.__version__}")
```

```
Transformers library imported successfully!
PyTorch version: 2.10.0+cpu
```

```
# Initialize a summarization pipeline with a pretrained model
# facebook/bart-large-cnn is a popular choice for abstractive summarization of longer texts.
# Due to an environment issue where 'summarization' task is not recognized by pipeline, we will load the model and tokenizer di
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

# We can still create a pipeline-like callable for convenience if needed, or use tokenizer/model directly
# summarizer = pipeline("summarization", model=model, tokenizer=tokenizer) # This would still fail due to the task name issue

# For now, we'll confirm direct loading
print("Tokenizer and Model for facebook/bart-large-cnn loaded successfully!")
```

```
vocab.json:        899k/?  [00:00<00:00,  32.0MB/s]

merges.txt:        456k/?  [00:00<00:00,  23.4MB/s]

tokenizer.json:        1.36M/?  [00:00<00:00,  38.5MB/s]

model.safetensors:  100%                                                    1.63G/1.63G  [00:28<00:00,  60.3MB/s]
Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights:  100%                                  511/511  [00:01<00:00,  552.57it/s,  Materializing param=model.encoder.layers.11.self_attn_layer_norm.weight]

generation_config.json:  100%                                                    363/363  [00:00<00:00,  14.0kB/s]
Tokenizer and Model for facebook/bart-large-cnn loaded successfully!
```

```
# Prepare diverse input texts for summarization
# We'll create a pandas DataFrame to store our sample texts along with their categories.

data = [
    {
        "category": "News Article",
        "text": "Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Ge
    },
    {
        "category": "Scientific Abstract",
        "text": "This paper presents a novel deep learning architecture for image segmentation using a combination of convoluti
    },
    {
        "category": "Product Review",
        "text": "I recently purchased the XYZ wireless earbuds, and I must say, I'm quite disappointed. The sound quality is me
    },
    {
        "category": "Conversational Text",
        "text": "Hey, did you catch the game last night? It was insane! The Lakers almost blew it in the fourth quarter, but Le
    },
    {
        "category": "Legal Document Excerpt",
        "text": "WHEREAS, the parties hereto desire to enter into this Agreement to define their respective rights and obligati
    }
]

df_texts = pd.DataFrame(data)
display(df_texts.head())

print(f"Prepared {len(df_texts)} texts from various domains.")
```

	category	text
0	News Article	Recent studies have shown a significant increa...
1	Scientific Abstract	This paper presents a novel deep learning arch...
2	Product Review	I recently purchased the XYZ wireless earbuds,...
3	Conversational Text	Hey, did you catch the game last night? It was...
4	Legal Document Excerpt	WHEREAS, the parties hereto desire to enter in...

```
Prepared 5 texts from various domains.
```

```
# Function to generate a summary for a given text, with customizable parameters
def generate_summary(text, max_length, min_length, do_sample=False, num_beams=4):
    # Tokenize the input text
    inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="pt")

    # Generate summary tokens
    summary_ids = model.generate(
        inputs["input_ids"],
```

```
        num_beams=num_beams,
        max_length=max_length,
        min_length=min_length,
        do_sample=do_sample
    )

    # Decode the generated tokens back to text
    summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
    return summary

# We will use the first text (News Article) for demonstrating different summary lengths
news_article_text = df_texts.loc[0, 'text']

print("Original News Article:")
print(news_article_text)
print("\n---")

# Generate a Short Summary
short_summary = generate_summary(news_article_text, max_length=50, min_length=10)
print("Short Summary (max_length=50, min_length=10):")
print(short_summary)
print("\n---")

# Generate a Medium Summary
medium_summary = generate_summary(news_article_text, max_length=100, min_length=30)
print("Medium Summary (max_length=100, min_length=30):")
print(medium_summary)
print("\n---")

# Generate a Long Summary
long_summary = generate_summary(news_article_text, max_length=150, min_length=50)
print("Long Summary (max_length=150, min_length=50):")
print(long_summary)
print("\n---")

# Generate a summary with do_sample=True (more creative/less deterministic)
sampled_summary = generate_summary(news_article_text, max_length=100, min_length=30, do_sample=True, num_beams=1) # num_beams=1
print("Sampled Summary (max_length=100, min_length=30, do_sample=True, num_beams=1):")
print(sampled_summary)
print("\n---")
```

```
Original News Article:
Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Germany and Sweden a

---
Short Summary (max_length=50, min_length=10):
Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Germany and Sweden a

---
Medium Summary (max_length=100, min_length=30):
Recent studies have shown a significant increase in renewable energy adoption across Europe. Experts predict that by 2030, a sub

---
Long Summary (max_length=150, min_length=50):
Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Germany and Sweden a

---
Sampled Summary (max_length=100, min_length=30, do_sample=True, num_beams=1):
Germany and Sweden are leading Europe's adoption of cleaner energy. Cost of renewables has decreased, so cost is decreasing. Exp

---
```

```
 summarized_texts = []

for index, row in df_texts.iterrows():
    category = row['category']
    original_text = row['text']

    # Generate a medium-length summary for consistency in comparison
    summary = generate_summary(original_text, max_length=100, min_length=30)

    summarized_texts.append({
        'category': category,
        'original_text': original_text,
        'generated_summary': summary
    })
```

```
df_summaries = pd.DataFrame(summarized_texts)
display(df_summaries)

print("Summaries generated for all diverse text types.")
```

	category	original_text	generated_summary
0	News Article	Recent studies have shown a significant increa...	Recent studies have shown a significant increa...
1	Scientific Abstract	This paper presents a novel deep learning arch...	'SegAttentionNet' achieves state-of-the-art pe...
2	Product Review	I recently purchased the XYZ wireless earbuds,...	The sound quality is mediocre at best, lacking...
3	Conversational Text	Hey, did you catch the game last night? It was...	The Lakers almost blew it in the fourth quarte...
4	Legal Document Excerpt	WHEREAS, the parties hereto desire to enter in...	The parties hereto desire to enter into this A...

```
Summaries generated for all diverse text types.
```

```
long_text = """The industrial revolution, which began in Great Britain in the late 18th century, marked a profound shift in hum

print("Original Long Text (first 200 characters):")
print(long_text[:200] + "...")
print(f"Total length of original text: {len(long_text.split())} words")

# Summarize the long text with a generous max_length to capture more information
long_text_summary = generate_summary(long_text, max_length=200, min_length=50)

print("\n--- Generated Summary ---")
print(long_text_summary)
print(f"Total length of summary: {len(long_text_summary.split())} words")
```

```
Original Long Text (first 200 characters):
The industrial revolution, which began in Great Britain in the late 18th century, marked a profound shift in human history, fund
Total length of original text: 314 words

--- Generated Summary ---
The industrial revolution began in Great Britain in the late 18th century. It was characterized by the transition from agrarian
Total length of summary: 48 words
```